

PROOF OF A SQUARE-ENDPOINT SUPERCONGRUENCE FOR CENTRAL-BINOMIAL CUBES

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ABSTRACT. We prove a conjectured square-endpoint supercongruence for sums of cubes of central binomial coefficients. If p is a prime with $p \equiv 3, 5, 6 \pmod{7}$, then

$$\sum_{k=0}^{p^2} \binom{2k}{k}^3 \equiv 8 + p^2 \pmod{p^3}.$$

The proof first converts the truncated sum to the square of a Legendre polynomial by a terminating Clausen identity and an exact carry analysis. A two-digit Bernstein expansion then evaluates that Legendre polynomial modulo p^2 ; its sole surviving coupled term is normalized by a polynomial Wronskian identity. Two previously known congruences of Z.-H. Sun provide the half-range vanishing and the trace-zero Legendre root. Exact finite computations and a separately implemented verifier were used only to attack signs, endpoints, and transcription errors; they are not substitutes for the universal argument. No assertion is made for split residue classes, prime powers, the prime 7, or modulus p^4 .

1. INTRODUCTION

Put

$$A(n) = \sum_{k=0}^n \binom{2k}{k}^3.$$

The square-endpoint congruence proved here appears as a conjecture in the record for OEIS A079727 [1]. The two external ingredients we use are Theorems 2.5 and 3.2 of Sun [2]; neither theorem already contains the square-endpoint conclusion. The first supplies a root of a Legendre polynomial modulo p , and the second supplies a half-range central-binomial sum modulo p^2 .

The new step is a lift across two base- p digits. Set

$$h = \frac{p-1}{2}, \quad N = \frac{p^2-1}{2}, \quad z^2 = -63.$$

The argument has two bridges. First,

$$A(p^2) - 8 \equiv P_N(z)^2 \pmod{p^3}.$$

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Second, whenever $(-63/p) = -1$ and $P_h(z) = 0$ modulo p ,

$$P_N(z) \equiv (-1)^h p \pmod{p^2}.$$

The second bridge is the essential lift. A direct two-digit expansion seems at first to contain several unrelated first-order terms. At the trace-zero root all uncoupled terms contain a factor $P_h(z)$ and vanish. The remaining term factors into a Bernstein derivative and a finite companion polynomial; their product is forced by a Bezout–Wronskian identity.

All congruences below that involve z are interpreted in the finite free ring

$$R_e = (\mathbb{Z}/p^e\mathbb{Z})[z]/(z^2 + 63).$$

For the relevant primes $p \geq 5$, the reduction R_1 is a field. Thus every division used below is by an explicitly identified unit.

2. PRELIMINARIES

Lemma 2.1 (Quadratic character and units). *Let $p \geq 5$ be prime and $p \equiv 3, 5, 6 \pmod{7}$. Then*

$$\left(\frac{-63}{p}\right) = \left(\frac{-7}{p}\right) = \left(\frac{p}{7}\right) = -1.$$

Consequently R_1 is a field. If

$$u = \frac{z+1}{2}, \quad v = \frac{z-1}{2},$$

then $uv = -16$, so z, u, v are units, and

$$z^p = -z, \quad u^p = -v, \quad v^p = -u \quad \text{in } R_1.$$

Proof. The factor 9 is a square. Quadratic reciprocity gives $(-7/p) = (p/7)$ for odd $p \neq 7$. The nonzero quadratic nonresidues modulo 7 are 3, 5, 6, proving the character assertion. Irreducibility of $z^2 + 63$ over $\mathbb{F}_p$ follows. The identities for u, v follow from $z^2 = -63$, and Frobenius sends z to the other root $-z$. $\square$

Lemma 2.2 (The endpoint central binomial coefficient). *For every prime $p \geq 5$,*

$$\binom{2p^2}{p^2} \equiv 2 \pmod{p^3}, \quad \left(\binom{2p^2}{p^2}\right)^3 \equiv 8 \pmod{p^3}.$$

Proof. For $m = p^2$ write

$$\frac{1}{2} \binom{2m}{m} = \prod_{j=1}^{m-1} \left(1 + \frac{m}{j}\right).$$

The factors with $p \nmid j$ have product $1 + p^2 \sum_{p \nmid j} j^{-1}$ modulo p^3 . The sum is 0 modulo p by grouping the units according to their residues modulo p . The factors with $j = p\ell$ give

$$\prod_{\ell=1}^{p-1} \left(1 + \frac{p}{\ell}\right).$$

Pairing ℓ with $p - \ell$ shows $H_{p-1} \equiv 0 \pmod{p^2}$, because $\sum_{\ell=1}^{p-1} \ell^{-2} \equiv 0 \pmod{p}$ for $p \geq 5$. Expanding the last product through order p^2 therefore gives 1 modulo p^3 . This proves the first congruence, and cubing proves the second. $\square$

3. THE TERMINATING CLAUSEN BRIDGE

Lemma 3.1 (Clausen bridge). *For every prime $p \geq 5$ with $p \equiv 3, 5, 6 \pmod{7}$,*

$$A(p^2) - 8 \equiv P_N(z)^2 \pmod{p^3} \quad \text{in } R_3.$$

Proof. The terminating Clausen identity gives, exactly over the rationals,

$$P_N(z)^2 = {}_3F_2\left(-N, N+1, \frac{1}{2}; 1, 1; 1-z^2\right).$$

Since $1 - z^2 = 64$ and $2N + 1 = p^2$, comparison of consecutive coefficients gives

$$(3.1) \quad P_N(z)^2 = \sum_{k=0}^N t_k R_k, \quad t_k = \binom{2k}{k}^3, \quad R_k = \prod_{j=0}^{k-1} \left(1 - \frac{p^4}{(2j+1)^2}\right).$$

Indeed the ratio of consecutive hypergeometric coefficients is

$$\frac{8(2k+1)((2k+1)^2 - p^4)}{(k+1)^3},$$

whereas $t_{k+1}/t_k = 8(2k+1)^3/(k+1)^3$.

The largest odd denominator in (3.1) is $2N - 1 = p^2 - 2$; no such denominator is divisible by p^2 . If $p \nmid 2j + 1$, the corresponding factor is 1 modulo p^3 . If $2j + 1 = p(2r + 1)$, it is $1 - p^2/(2r + 1)^2$, which is still p -integral. Thus no negative valuation has been hidden in (3.1).

Write $k = ap + b$ with $0 \leq a, b < p$. Kummer's theorem gives

$$(3.2) \quad v_p \binom{2k}{k} = 0 \iff 0 \leq a, b \leq h.$$

Outside this no-carry rectangle, t_k is divisible by p^3 . Its largest index is $hp + h = N$, so the rectangle lies exactly within the terminating range and all other terms may be discarded modulo p^3 in both the Clausen sum and $A(p^2 - 1)$.

For $k = ap + b$ in the rectangle, the odd multiples of p below $2k$ are $p(2r + 1)$ for $0 \leq r < a$. Hence

$$(3.3) \quad R_{ap+b} \equiv 1 - p^2 E_a \pmod{p^3}, \quad E_a = \sum_{r=0}^{a-1} \frac{1}{(2r+1)^2} \pmod{p}.$$

Lucas' theorem gives $t_{ap+b} \equiv t_a t_b \pmod{p}$. The complete correction from (3.3) is therefore

$$(3.4) \quad -p^2 \left(\sum_{a=0}^h t_a E_a \right) \left(\sum_{b=0}^h t_b \right) \pmod{p^3}.$$

Sun's Theorem 3.2 states, in precisely these residue classes, that the second factor in (3.4) is 0 modulo p^2 [2, Theorem 3.2]. Thus $P_N(z)^2 \equiv A(p^2 - 1) \pmod{p^3}$. Adding the endpoint from Lemma 2.2 proves the claim. $\square$

4. A TWO-DIGIT LEGENDRE LIFT

We next isolate the lift that turns the trace-zero root at degree h into a first-order value at degree N .

Lemma 4.1 (Digit-binomial expansion). *For $0 \leq a, b \leq h$,*

$$(4.1) \quad \binom{ph+h}{pa+b} \equiv \binom{h}{a} \binom{h}{b} (1 + pE_{a,b}) \pmod{p^2},$$

where

$$(4.2) \quad E_{a,b} = hH_h - aH_b - (h-a)H_{h-b}$$

$$(4.3) \quad = h(H_h - H_{h-b}) + a(H_{h-b} - H_b).$$

Proof. All factorials below have length less than p , so their factors are units. The identity

$$\binom{ph+h}{pa+b} = \binom{ph}{pa} \frac{\prod_{i=1}^h (ph+i)}{\prod_{i=1}^b (pa+i) \prod_{i=1}^{h-b} (p(h-a)+i)}$$

combined with

$$\prod_{i=1}^r (i + pX) \equiv r!(1 + pXH_r) \pmod{p^2}$$

gives (4.1)–(4.2); the standard unit-block cancellation $\binom{ph}{pa} \equiv \binom{h}{a} \pmod{p^2}$ is more than sufficient here. Rearrangement gives (4.3). $\square$

Define the finite companion polynomial

$$W_h(z) = \sum_{j=1}^h \frac{1}{j} P_{j-1}(z) P_{h-j}(z).$$

Lemma 4.2 (Bernstein companion and Wronskian). *With $u = (z+1)/2$ and $v = (z-1)/2$,*

$$(4.4) \quad \sum_{b=0}^h \binom{h}{b}^2 u^b v^{h-b} (H_{h-b} - H_b) = -W_h(z).$$

Moreover

$$(4.5) \quad (1 - z^2)(P'_h W_h - P_h W'_h) = 1 - P_h^2.$$

Proof. Let D_h denote the left side of (4.4). Pascal's identity and $H_{m+1} = H_m + 1/(m+1)$ give

$$(4.6) \quad (h+1)D_{h+1} = (2h+1)zD_h - hD_{h-1}, \quad D_0 = 0, \quad D_1 = -1.$$

On the other hand, the Legendre generating function gives

$$\sum_{h \geq 0} W_h(z)t^h = (1 - 2zt + t^2)^{-1/2} \int_0^t (1 - 2zs + s^2)^{-1/2} ds.$$

Coefficient extraction yields the same recurrence as (4.6), but with $W_0 = 0, W_1 = 1$. Hence $D_h = -W_h$.

For completeness, the finite second solution

$$Q_h = \frac{1}{2}P_h \log \frac{1+z}{1-z} - W_h$$

of the Legendre differential equation satisfies $P_h Q'_h - P'_h Q_h = 1/(1-z^2)$. Substitution and simplification give (4.5), which is a polynomial identity and so remains valid after reduction modulo p . $\square$

Lemma 4.3 (Legendre–Lucas lift). *Let $p \geq 5$ be prime, suppose $(-63/p) = -1$, and suppose $P_h(z) = 0$ in R_1 . Then*

$$(4.7) \quad P_N(z) \equiv (-1)^h p \pmod{p^2} \quad \text{in } R_2.$$

Proof. The exact Bernstein form is

$$(4.8) \quad P_N(z) = \sum_{k=0}^N \binom{N}{k}^2 u^k v^{N-k}.$$

By Lucas' theorem, a term can survive modulo p^2 only when $k = pa + b$ with $0 \leq a, b \leq h$.

Lemma 2.1 and the fact that u, v are units give uniquely defined $\alpha, \beta \in R_1$ such that, for arbitrary lifts to R_2 ,

$$(4.9) \quad u^p = -v(1 + p\alpha), \quad v^p = -u(1 + p\beta).$$

Put

$$A_a = \binom{h}{a}^2 u^{h-a} v^a, \quad B_b = \binom{h}{b}^2 u^b v^{h-b}.$$

Inserting Lemma 4.1 and (4.9) into (4.8), including every first-order term, yields

$$(4.10) \quad P_N(z) \equiv (-1)^h \sum_{a,b} A_a B_b \{1 + p[2E_{a,b} + a\alpha + (h-a)\beta]\} \pmod{p^2}.$$

Both unweighted channel sums are $P_h(z)$. Therefore the constant term, the α and β terms, and the first term in (4.3) vanish modulo p^2 : each has an unweighted $P_h(z)$ factor, with an additional outside p in every first-order term. The only surviving term is

$$(4.11) \quad P_N(z) \equiv (-1)^h 2p \left(\sum_{a=0}^h a A_a \right) \left(\sum_{b=0}^h B_b (H_{h-b} - H_b) \right) \pmod{p^2}.$$

Let $C_1 = \sum_b bB_b$. Reversing $a \mapsto h - a$ gives $\sum_a aA_a = hP_h - C_1 = -C_1$ at the root. Differentiating the Bernstein form for P_h and using $uv = -16$ and $v - u = -1$ gives

$$P'_h(z) = C_1 \left(\frac{1}{2u} - \frac{1}{2v} \right) = \frac{C_1}{32}.$$

Thus the first factor in (4.11) is $-32P'_h(z)$; by Lemma 4.2, the second is $-W_h(z)$. At $P_h(z) = 0$, (4.5) gives

$$P'_h(z)W_h(z) = \frac{1}{1 - z^2} = \frac{1}{64}.$$

The scalar in (4.11) is consequently $2(-32P'_h)(-W_h) = 1$, proving (4.7). Notice that (4.5) also proves that the root is simple, so no hidden division by zero occurs. $\square$

5. MAIN RESULT

Theorem 5.1. *For every rational prime $p \equiv 3, 5, 6 \pmod{7}$, $\sum_{k=0}^{p^2} \binom{2k}{k}^3 \equiv 8 + p^2 \pmod{p^3}$.*

Plain-language endpoint. For every rational prime p congruent to 3, 5, or 6 modulo 7, the sum from $k=0$ through p squared of the cube of the central binomial coefficient is congruent to 8 plus p squared modulo p cubed.

Proof. First suppose $p \geq 5$. Sun's Theorem 2.5 gives $P_h(\sqrt{-63}) \equiv 0 \pmod{p}$ for $p \equiv 3, 5, 6 \pmod{7}$ [2, Theorem 2.5]; its displayed sign is immaterial. Lemmas 2.1 and 4.3 therefore give

$$P_N(z)^2 \equiv p^2 \pmod{p^3}.$$

Lemma 3.1 now yields $A(p^2) \equiv 8 + p^2 \pmod{p^3}$.

The remaining eligible prime is $p = 3$, where the quadratic algebra is ramified and the preceding lift is not used. Direct integer summation gives

$$A(9) = 117106008311825 \equiv 17 = 8 + 3^2 \pmod{27}.$$

This completes every residue class and endpoint in the statement. $\square$

6. CERTIFICATE OR COMPUTATIONAL VERIFICATION

The proof was discovered with exact symbolic and modular experiments, then audited in a fresh context. A self-contained verifier recomputes the Clausen coefficient ratio, Kummer support, all digit-binomial cases in a regression range, the companion and Wronskian polynomial identities, the Frobenius expansion, and direct values of the target congruence. It imports no owner research module. Thirty-five deliberately corrupted inputs or identities were rejected, including wrong endpoints, deleted residue hypotheses, changed Frobenius and Wronskian signs, and a false p^2 replacement for the p^4 Clausen factor.

These checks test implementation and transcription. The infinite prime quantifier is discharged only by Lemmas 3.1 and 4.3 together with the cited

Sun congruences. No numerical approximation, finite search cutoff, optimizer output, or model confidence is used as a proof step. No proof assistant was used.

7. EQUALITY CASES AND LIMITATIONS

The theorem proves precisely the inclusive sum from $k = 0$ through $k = p^2$ for rational primes in the three nonsquare residue classes modulo 7. It does not claim the same formula for $p \equiv 1, 2, 4 \pmod{7}$, for composite moduli or prime powers substituted for p , for $p = 7$, for a different upper endpoint, or modulo p^4 . The trace-zero hypothesis in the lift is essential to the channel cancellations used in (4.10). The proof also does not promote any of the stronger empirical patterns recorded during discovery.

DATA, CODE, AND COMPUTATIONAL DISCLOSURE

The project archive contains the formal statement, proof dependency graph, exact modular data, an owner verifier, a separately implemented independent verifier, mutation tests, and the full audit reports. The principal replay commands are documented with those artifacts. All decisive arithmetic uses integers, rational functions with explicit unit denominators, finite-field extensions, or symbolic polynomial identities.

AI-assisted tools contributed to route exploration, algebraic expansion, test generation, manuscript drafting, and adversarial review. The final claim was accepted only after an independent reconstruction of the proof, primary-source endpoint checks, and fail-closed exact verification. Human readers should regard the displayed universal derivation, not the finite experiments, as the proof.

REFERENCES

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